

MetAct: Fabrication Technique for Interactive 3D Objects Using Low-Melting-Point Metal Extrusion on Standard FDM 3D Printers

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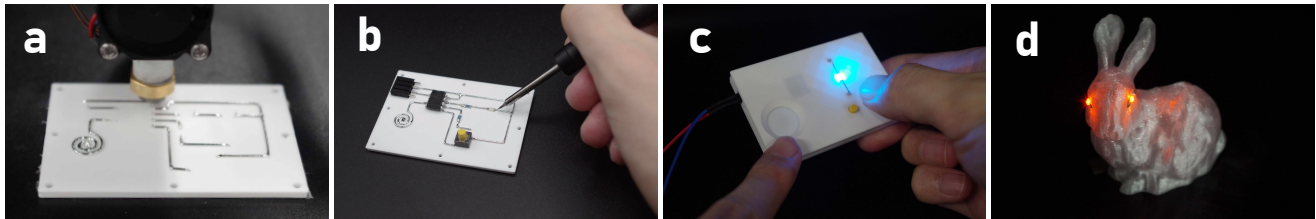


Figure 1: (a) Direct drawing of circuit trace on 3D printed substrate. (b) Manual soldering of the solder traces and electronic components. (c) Two-layer electronic circuit. (d) Stanford Bunny with embedded wiring and LEDs.

Abstract

We present MetAct, an accessible fabrication technique that enables interactive 3D objects by directly extruding low-melting-point metal using a FDM 3D printer. Unlike existing approaches that rely on conductive filaments, ink-based processes, or custom hardware, MetAct allows highly conductive and solderable metal traces to be drawn or embedded directly on 3D printed objects using a standard, consumer-grade, single-extruder FDM 3D printer. Our evaluation shows that the technique can be applied to a variety of materials while providing sufficient adhesive strength for rapid prototyping. We demonstrate several prototypes, including custom multi-layer electronic circuits, objects that activate simply by being placed on a powered surface, flexible objects whose shape-retention properties are enhanced by printed metal, and metal-only structures fabricated with an FDM printer. By allowing users to modify conductive traces directly during fabrication, MetAct shortens the gap between designing interactions and building working physical prototypes.

CCS Concepts

• Human-centered computing → Interactive systems and tools.

Keywords

Low-melting-point metal, Interactive 3D object, Electronic circuit, 3D printer

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1 INTRODUCTION

Digital fabrication tools such as FDM 3D printers have become widely accessible, enabling individuals to fabricate customized objects at home. As a result, personal fabrication has expanded beyond shape-making toward supporting creativity, experimentation, and everyday problem solving. However, most home-fabricated artifacts remain static physical objects whose functionality is limited to geometry alone.

Interactive functionality is essential for fabricated objects to participate in everyday use. While embedding electronic circuits is a common solution, existing approaches often rely on external development boards or custom PCBs. These solutions introduce constraints in size, geometry, and fabrication turnaround time, which hinder rapid iteration during prototyping.

Although prior research has explored rapid circuit fabrication techniques, several challenges remain unresolved. Conductive filaments typically suffer from low conductivity, ink-based techniques often require flat substrates or specialized equipment, and robust bonding of electronic components remains difficult. More importantly, many existing techniques interrupt the immediacy of personal fabrication by requiring multi-step workflows or advance planning of circuit layouts.

In this paper, we present MetAct, an accessible fabrication technique that enables interactive 3D objects by directly extruding low-melting-point metal using a standard, consumer-grade, single-extruder FDM 3D printer. MetAct allows users to draw or embed highly conductive and solderable metal traces directly onto 3D printed objects, supporting rapid, hands-on exploration of interactive artifacts. Through experiments and a set of interactive prototypes, we demonstrate how MetAct enables on-the-fly fabrication of functional electronic structures that support interactive behaviors.



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2 RELATED WORK

2.1 Fabricating Interactive 3D Objects

Embedding electronic circuits into 3D objects commonly involves fabricating the object and the printed circuit board (PCB) separately, followed by manual assembly. To lower the barrier of PCB fabrication, many rapid prototyping techniques have been proposed. Instant Inkjet Circuits [8] enable fast circuit fabrication using conductive ink and standard inkjet printers, but are limited to flat, paper-based substrates. Similarly, other rapid circuit fabrication techniques [15] assume planar surfaces and do not directly support integration with 3D printed geometries.

Alternative approaches fabricate circuits directly onto 3D printed objects. Techniques using conductive filaments [4, 6, 13, 14, 16] allow circuit fabrication with standard dual-extruder printers, but suffer from low conductivity and material contamination. Conductive ink dispensing [2, 3, 10, 18] achieves higher conductivity but requires customized hardware or additional fabrication steps.

User-assisted techniques such as spraying conductive ink [12], screen printing [19], hydroprinting [5], copper tape application [17], and hot stamping [7] provide flexible alternatives, but rely heavily on manual intervention. In contrast, MetAct emphasizes immediacy and accessibility by enabling metal-based circuit fabrication within a standard FDM printing workflow.

2.2 Extruding Low-Melting-Point Alloys

Low-melting-point alloys have been explored for fabricating conductive structures using 3D printers [1, 11, 20]. These approaches offer high conductivity and solderability but often rely on casting molten metal into pre-designed grooves, requiring circuit layouts to be finalized before object fabrication. Recent work has proposed tension-based metal printing [9], but requires specialized hardware.

MetAct differs by enabling direct metal extrusion using widely available FDM printers, prioritizing rapid experimentation and on-demand circuit creation.

3 PROPOSED TECHNIQUE

MetAct enables users to fabricate interactive 3D objects by extruding low-melting-point metal using a standard single-extruder FDM 3D printer. A core design goal is to preserve the accessibility and immediacy of home fabrication while introducing highly conductive and solderable wiring.

Although replacing thermoplastic filament with metal may appear straightforward, several technical challenges arise. Low-melting-point metals undergo phase transitions during extrusion, leading to significant viscosity changes. Adhesion between metal traces and thermoplastic substrates is also critical. Furthermore, previous studies report severe nozzle degradation when extruding metal using standard brass nozzles.

To address these issues, we explore printing parameters, evaluate adhesion strength across common materials, and replace the standard nozzle with a commercially available tungsten-carbide nozzle. This configuration enables stable metal extrusion while maintaining printer durability.

MetAct provides three key benefits: (1) high electrical conductivity, (2) direct solderability of printed traces, and (3) transformation of a standard FDM printer into a lightweight metal fabrication tool.

The fabrication workflow consists of three steps. First, a base object is printed using thermoplastic filament. Second, the printer extrudes metal along a designed path to form conductive traces on or within the object. Third, electronic components are mounted directly onto the printed traces via manual soldering. Both surface wiring and embedded wiring using internal channels are supported.

4 EVALUATION

We evaluated MetAct in terms of trace quality and mechanical adhesion to confirm its feasibility. We used an Ender 3 V3 KE (Creality), a tin filament with a diameter of 1.75 mm composed of 99% tin (Sn 99%), and a 0.6 mm tungsten-carbide nozzle.

4.1 Quality Evaluation of Direct-Draw Trace Printing

To clarify the applicability of the proposed technique and the effects of printing parameters, we conducted a comparative photo-based evaluation of direct-draw trace quality. The observation criteria were (i) line continuity and (ii) line-width uniformity. We show how the extrusion amount, defined as the ratio of the supplied filament length to the drawn line length, impacts trace quality.

For the tin filament, we adopted empirically determined baseline parameters that prevent line breakage and stabilize line width: a nozzle temperature of 260°C, a nozzle-to-build-plate distance of 0.55 mm, a head speed of 300 mm/min, and an extrusion amount of 0.14 mm/mm. To investigate the effect of extrusion amount, we varied it from 0.04 mm/mm to 0.49 mm/mm in increments of 0.05 mm/mm. On each substrate, we printed ten straight lines of 25 mm length while stepwise changing the extrusion amount. We repeated this procedure three times for each of five substrates: the build plate, PLA, ABS, PETG, and TPU.

To compare the effects of parameter changes between metal and regular filament, gray PLA filament was printed on the same substrates. For PLA, the nozzle temperature was set to 210°C and the Z-offset to approximately 0.3 mm. The extrusion amount and head speed were kept identical across materials. The build-plate temperature was fixed at 30°C for all conditions. Note that different printers were used to fabricate the substrates to accelerate the evaluation process.

The results indicate that the tin filament was printable on all tested substrates. For PLA filament, the line width increased as the extrusion amount increased while maintaining overall print quality. In contrast, metal traces exhibited weaker dependence of line width on extrusion amount. Instead, swelling at the starting point of traces was observed at higher extrusion amounts. On PLA and PETG substrates, this swelling appeared from 0.24 mm/mm, and from 0.34 mm/mm it was observed on all plastic substrates. These results suggest that extrusion amount is a critical parameter for the proposed technique and that precise control of line width remains challenging in the current setup.

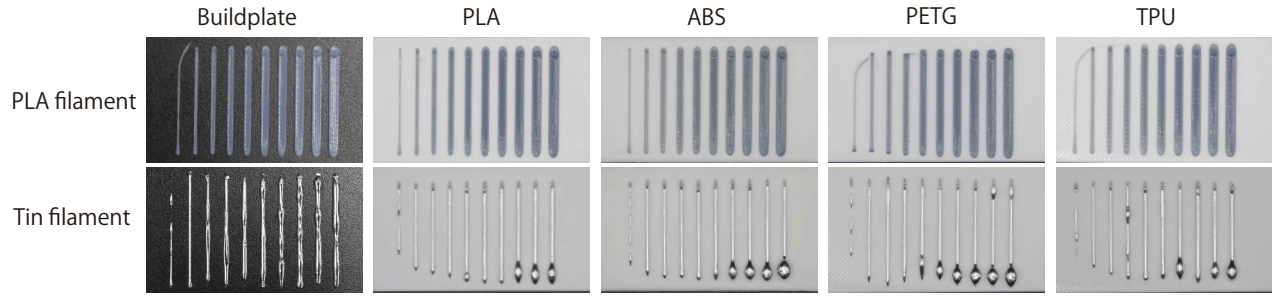


Figure 2: Comparison of direct-draw trace quality across substrates and materials. While the width of PLA traces increases with extrusion amount, the width of metal traces remains nearly constant and instead exhibits swelling at the beginning of the printed traces.

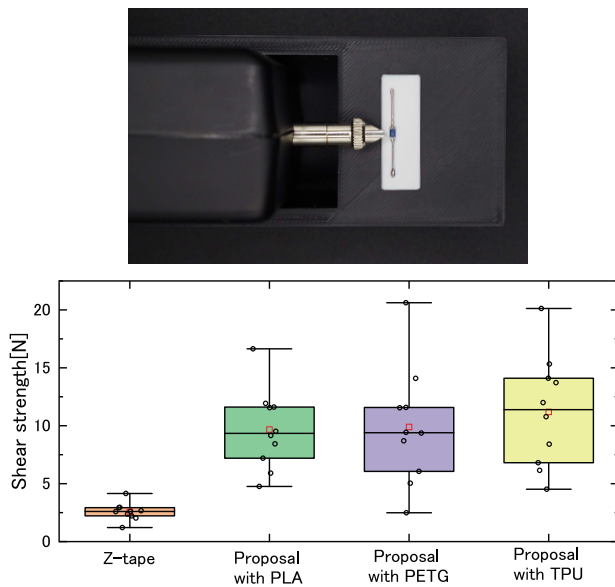


Figure 3: Setup for the shear strength evaluation (top) and graph of measured shear strength of chip resistors mounted on different substrate materials (bottom). ABS substrates exhibit very low adhesion and are therefore excluded from the graph, whereas other substrates show higher shear strength than Z-tape.

4.2 Mechanical Strength of Mounted Circuits

Adhesion strength between printed traces and substrates is a key concern for practical electronic circuits. To evaluate this, we measured the shear strength required to detach a mounted electronic component. We used a 3216 (metric) chip resistor because applying force directly to a trace often causes bending rather than detachment.

For each substrate material (PLA, ABS, PETG, and TPU), we prepared a flat substrate of 30 mm × 10 mm fabricated by 3D printing. Two 10 mm tin traces were printed with a spacing of 3.2 mm and a target line width of 1 mm or less. A chip resistor was manually soldered onto the traces, and shear force was applied using a force



Figure 4: (a) A 3D object with embedded conductive paths. The bunny's eyes function as touch sensors, demonstrating that the technique enables small, precise touch electrodes. (b) Lightweight 4D printing enabled by metal traces printed on a flexible TPU substrate. The shape is manually deformed and retained after the applied force is released. (c) A metal Stanford Bunny fabricated by stacking extruded metal traces (59 layers at a 0.5 mm layer pitch). This example illustrates that the technique enables metal fabrication using a standard FDM 3D printer.

gauge. This procedure was repeated ten times for each substrate. As a baseline, we mounted the same resistor on a PLA substrate using Z-tape.

The results show that adhesion strength depends strongly on substrate material. On ABS substrates, traces frequently detached during soldering and are therefore omitted from the figure. In contrast, traces printed on PLA, PETG, and TPU generally exhibited higher shear strength than Z-tape. These results indicate that the proposed technique is suitable for rapid prototyping and offers flexibility in substrate material selection.

5 APPLICATIONS

After confirming the feasibility of MetAct through evaluations, we explore its application space using a set of representative prototypes that illustrate how interaction behaviors can be directly shaped through fabrication.

First, to examine the accuracy of direct metal drawing, we fabricated a two-layer circuit board by extruding metal into vias designed in the second substrate layer. The circuit includes a 2.54 mm-pitch PIC microcontroller, two surface-mount LEDs with resistors, a pin header, a tactile switch, and a touch sensor (Fig. 1(c)). The fabrication process consists of three steps: (1) printing metal traces on the first substrate layer after its fabrication (Fig. 1(a)); (2) manually mounting and soldering electronic components (Fig. 1(b)); and (3)

stacking a second insulating substrate layer and extruding metal into pre-designed vias to form inter-layer connections and traces. The results confirm that conductive traces can be fabricated without short circuits even for surface-mount components. This demonstrates that practical electronic devices with moderate complexity can be rapidly prototyped using inexpensive, single-extruder FDM printers. The resulting multi-layer structure, achieved through simple manual intervention, also improves robustness against mechanical stress and enables functional layering, such as electrical insulation for the touch sensor and visual differentiation using layered substrates. In this prototype, one LED is mounted on the first layer, and its emitted light is modified by the overlaid second substrate.

To investigate wiring integration within 3D geometries, we fabricated an interactive object by injecting metal into internal channels designed inside a Stanford Bunny (Fig. 1(d)). The embedded wiring allows LEDs mounted in the eyes to activate when the object is placed on a powered surface. This “reacts-by-placement” behavior highlights a key characteristic of MetAct: interaction behaviors can be embedded directly into the physical structure of an object during fabrication. Another example is shown in Fig. 4(a), which is similar to Fig. 1(d) but is used as a touch electrode. MetAct enables the fabrication of small, precise touch electrodes directly on 3D objects, supporting fine-grained touch interaction without additional components.

Beyond electrical connectivity, MetAct can also serve as a structural element. By directly printing metal traces onto a flexible TPU substrate, we created an object that exhibits shape-retention along specific directions (Fig. 4(b)). This suggests a lightweight form of 4D printing, where shape retention emerges from material interaction, enabling designers to control physical behavior through fabrication rather than post-assembly mechanisms.

Finally, by stacking extruded metal traces, we fabricated a Stanford Bunny composed solely of tin filament by stacking 59 layers at a layer pitch of 0.5 mm (Fig. 4(c)). This example illustrates that MetAct effectively transforms a standard FDM printer into a simple metal printer. Due to the softness of low-melting-point tin at room temperature, the resulting objects are flexible rather than rigid, and the current implementation is limited to shell-like geometries, leaving some surfaces open (e.g., the back of the bunny) and preventing the fabrication of horizontally oriented, flat plate-like structures.

6 LIMITATIONS

While we have demonstrated the feasibility of MetAct, several limitations remain. First, impurities can accumulate inside the nozzle during metal extrusion, which may cause unstable extrusion in long-duration prints. As a result, regular nozzle cleaning is currently required to maintain print quality, limiting long-term autonomous operation. Second, precise control of trace width remains challenging, and the current technique also limits the fabrication of flat, plate-like structures. At present, we do not have a reliable method to address these issues; however, overcoming them could significantly expand the application space of the technique. Third, the generation of G-code for metal extrusion is currently performed manually. Automated G-code generation tailored to metal extrusion would improve usability and reproducibility. Finally, although MetAct is

designed to work with simple single-extruder FDM printers, using a dual-extruder printer could further speed up the prototyping process and broaden the range of possible applications.

7 CONCLUSION

We presented MetAct, an accessible fabrication technique that enables interactive 3D objects by extruding low-melting-point metal using widely available consumer-grade FDM 3D printers. Our evaluations show that MetAct can be applied to a variety of surface materials while providing sufficient adhesion of printed metal traces for prototyping. Through a set of representative prototypes, we demonstrated that the technique supports a broad application space for interaction design, ranging from custom electronic circuits to structurally augmented objects and exploratory forms of 4D printing, and can even repurpose a standard FDM printer as a simple metal fabrication tool.

Acknowledgments

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